

EME 5111 - ENGINEERING ECONOMICS ASSIGNMENT

1. What are the different methods for comparing engineering project alternatives?
2. What factors determine whether to replace an asset?
3. How does the Capital Recovery Method help in replacement analysis?
4. A machine costs £2,00,000 and has an annual maintenance cost of £30,000. If a new machine costs £3,50,000 with a maintenance cost of £10,000 per year, which machine is more economical over 10 years?
5. What is the Time Value of Money?
6. What is Engineering Economics?
7. What are the key objectives of economic decision-making in engineering
8. What methods are used to handle uncertainty in engineering economic decisions?
9. Explain the Expected Value Approach with an example.
10. What are the different methods of depreciation?
11. A machine costs £1,00,000, has a salvage value of £10,000, and a life of 5 years. What is the annual depreciation?
12. What are the different factors influencing material selection in engineering projects?
13. How can design selection impact the economic feasibility of a product?
14. Give an example of how material selection affects project cost
15. A steel component costs £500 per unit with a lifecycle of 10 years, while an aluminium component costs £800 per unit but lasts for 15 years. Which material is more economical per year
16. Explain the concept of "Economic Life of an Asset.
17. How do you calculate the future value of an investment?
18. If an investment of £50,000 is made at 12% annual interest for 5 years, what will be the future value?